
PalmSource™ 99
Get Your Hands on TRG's Enterprise Deployment Tools
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Introduction

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- Welcome to the TRGpro developer's session
- Brief history of TRG
- Evolution of TRGpro
- Overview of the TRGpro
  - Expansion slot comparison
  - TRGpro hardware
  - TRGpro software
- TRG Enterprise Deployment Tools

Brief History of TRG

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- TRG started as a electronics and software design consulting firm in 1991
- We have been involved in dozens of design projects for numerous clients over the years
- Clients include Rockwell/Collins, Intermec, KONE, Schlumberger, 3M, 3COM, OpenSky and others
- Core design team has been working on handheld computers since 1989 with the introduction of the DFM TravelLite tablet computer

Evolution of TRGpro

- TRG has been a leader in Palm Computing® platform add-ons since the platform was introduced in 1996
  - First to produce internal memory upgrade
  - First to add Flash
  - First to provide 8 MB upgrade
  - First to enable Palm V memory expansion through FlashPro
- Recognized customer's needs
  - Enhanced storage options
  - Improved, louder audio
  - Additional features via easily accessible add-on cards

Expansion Slot Comparison

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- Several expansion slot design options were considered
  - PC Card (PCMCIA)
  - Smart Media Card
  - MultiMedia Card
  - Proprietary Card
  - CompactFlash (CF)

Expansion Slot Comparison

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- PC Card (PCMCIA)
  - +Mature industry standard
  - +Variety of vendors and functions to choose from
  - +Cards can be interchanged with other systems
  - +Supports FAT File System (FFS)
  - +Supports execute in place (XIP)
  - -Too big for Palm form factor

Expansion Slot Comparison

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- Smart Media Card
  - +Cards can be interchanged with other systems
  - +Gaining industry acceptance
  - +Very small and light
  - +Low power
  - -Too small to support I/O functionality
  - -Relatively new type of media
  - -Does not provide ATA interface (burden on host)
  - -Does not support execute in place (XIP)

Expansion Slot Comparison

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- MultiMedia Card
  - +Cards can be interchanged with other systems
  - +Extremely small and light
  - +Low power
  - -Too small to support I/O functionality
  - -Relatively new type of media
  - -Does not provide ATA interface (burden on host)
  - -Does not support execute in place (XIP)
  - -Serial interface

Expansion Slot Comparison

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- Proprietary Card
  - +Good integration with Palm OS
  - +Good versatility
  - +Supports execute in place (XIP)
  - -Not an industry standard
  - -Can not interchange with other systems
  - -Card vendors bear the burden of a single relatively limited market

Expansion Slot Comparison

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- CompactFlash (CF)
  - +Mature industry standard (CFA founded in 1995, 143 member companies including TRG and 3COM)
  - +Growing list of vendors and functions
  - +Cards can be interchanged with other systems
  - +Small enough to fit Palm form factor
  - +Contains ATA interface to facilitate use of FAT File System (FFS) storage cards
  - -Does not support execute in place (XIP)

Expansion Slot Conclusion

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- Compact Flash was the obvious choice
  - PC Card—Too big
  - Smart Media—No versatility
  - Proprietary—Not industry standard (ala MicroChannel, BetaMax)
- Addition of CF does not significantly change Palm IIIx form factor
- Integrates very well with existing PC and Macintosh systems

TRGpro Hardware Overview

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- Standard TRGpro hardware configuration
  - Internal Flash memory 2 MB standard
  - System memory 8 MB
  - Type II CF+ power level 0 slot
  - Enhanced Audio
  - Cradle
- Customizations for enterprise customers
  - Different memory configurations
  - Custom Flash loads
  - Enhanced screen options
  - Private labeling

TRGpro Handheld Computer

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- Industry standard expansion, CompactFlash
- Increased memory
- On board Flash
- Flexible RAM/ROM configurations
- Voice quality sound output
- Radio ready design
  - 930 MHz (Pagers)
  - 2.4 GHz (Bluetooth)
- Palm Computing® Platform Compatible



TRGpro Software Overview

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- TRG OS extension APIs
  - CF hardware interface
  - FAT File System (FFS)
  - Enhanced audio
- CFpro
  - Application for browsing CF storage cards
  - Moves and copies files and databases
- CFBackup
  - Backs up entire unit to CF
  - Backs up a fully loaded TRGpro in 45 seconds!
- CFRestore available in late 2000 ☺

TRG Enterprise Deployment Tools

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- IMAGEpro
  - Creates customized Flash image
  - Enables enterprise clients to create custom OS and Flash based applications
  - Easy deployment to workforce
- SFApro (Self Flashing Applications)
  - Automatic load into Flash
  - Replace obsolete version
  - Easy deployment to workforce



TRGpro Hardware Introduction

Doug DeVries
Lead Hardware Engineer

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Introduction

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- Palm IIIx™ hardware review
- TRGpro hardware description
- CF slot
- Enhanced audio
- Memory options
- Radio ready for wireless devices
- Battery life
- Customization possibilities

Palm IIIx™ Hardware Review

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- Motorola DragonBall-EZ
- 4 MB DRAM
- 2 MB Flash
- IrDA
- Serial port
- Touchscreen
- Piezoelectric speaker
- 160x160 FSTN LCD with EL backlight

TRGpro Hardware Description

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- Based on Palm IIIx design
- Added CF slot
- Increased DRAM to 8 MB
- Added support for more Flash options
- Enhanced audio
- Higher current power supply

CF Slot

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- CF cards weigh a half ounce and are the size of a matchbook
- They provide complete PCMCIA-ATA functionality and compatibility
- CF cards support both storage and I/O

CF Slot

- Buffers
  - Card can be safely inserted or removed at any time
  - Performs byte-swapping functions
  - Isolates CF from system during sleep
- Power switch
  - Connects 3.3 V power supply to CF card
  - Automatically disconnects CF card from power within micro-seconds of low battery condition
- CF Bus timing created by on board logic

Enhanced Audio

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- Speaker replaces piezoelectric element
  - Louder, better frequency response
- Audio amplifier added
  - Differential output stage to drive speaker
  - Built in 64 step software controlled attenuator
- Audio low-pass filter added
  - 5th order elliptical transfer function
  - 4 kHz corner frequency
  - Simple high-pass corner at 200 Hz
  - Removes PWM artifacts
- Enables voice quality audio and DTMF

Memory Options

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- 8 MB DRAM
  - Maximum directly supported by EZ
- 2 MB Flash standard
- Up to 16 MB Flash supported
  - Two sites
  - Combinations include 1, 2, 3, 4, 8, 9, 10 & 16 MB

Radio Ready for Wireless Devices

- RF emissions significantly reduced over entire band (30 MHz–1 GHz)
- ESD immunity improved over CE mark
- 929-932 MHz band (FLEX Pagers)
  - Emissions less than -87 dBm, 18 dBuV/m at 1 m
  - Maintains >6 dB margin in band
- 2.4-2.5 GHz band (Bluetooth)
  - Emissions less than -92 dBm, 13 dBuV/m at 1 m
  - Maintains >6 dB margin in band
- RF immunity >10 V/m

Battery Life

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- Complicated number to predict
  - ESR of batteries
  - Duty cycle of usage
  - Exactly how each piece of hardware is being used
- TRGpro with no CF behaves exactly as a Palm IIIx
- CF cards consume additional power
  - Typically well under 75 mA
- TRGpro supports up to 300 mA continuous, 500 mA peak delivered to CF slot

Customization Possibilities

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- TRGpro can be “modified”, and “cutomized”
- “Modifications” are chosen from a standard list
  - Different memory configurations
  - Custom Flash loads
  - Change sound options
  - Etc.
- “Customizations” are created according to specific customer requirements
  - Different form factors
  - Peripherals added to mother board
  - Etc.



TRGpro Software Introduction

Mike Walter
Lead Software Engineer

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Introduction

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- TRGpro software architecture
  - CompactFlash (CF)
  - FAT File System (FFS)
  - Enhanced Audio
- Development environment

TRGpro Software Architecture

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- Based on Palm OS® ver 3.3
- Enhancements implemented through shared libraries
- Hardware changes to standard features transparent to developers

CF API

- Provides interface to CF slot hardware
- Implemented as a shared library
- Encapsulates hardware interface
- Enables developers to create custom CF card drivers (pagers, ethernet, etc.)

CF API

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- Important functions
  - CfBaseAddress()
  - CfPowerSlot()
  - CfReset()
  - CfSetDataWidth()
  - CfSetSwapping()
  - CfCardInserted()
- Notifications
  - Applications register themselves at startup
  - Applications notified of CF insertion/removal via launch codes

FAT File System (FFS) API

- Provides FFS support functions for CF cards
- Implemented as a shared library
- Supports file and directory manipulation
- Interface based upon unbuffered file/disk library calls typically used in C

FAT File System (FFS) API

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- **FFS Prototype**

```
Int16 FfsOpen(UInt16  libRef,
              char   *path,
              Int16   flags,
              Int16   mode);
```

- **Standard C Prototype**

```
int Open(char *filename,
         int   oflag,
         int   pmode);
```

FAT File System (FFS) API

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- Important file access functions
  - FfsCreate()
  - FfsOpen()
  - FfsClose()
  - FfsWrite()
  - FfsRead()
  - FfsLseek()
- Important file management functions
  - FfsRemove()
  - FfsRename()
  - FfsStat()

FAT File System (FFS) API

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- Example directory management functions
  - FfsChdir()
  - FfsMkdir()
  - FfsRmdir()
  - FfsFindfirst()
  - FfsFindnext()
- Example disk management functions
  - FfsFormat()
  - FfsFlushdisk()
  - FfsCardIsATA()

Audio API

- Provides support functions for TRGpro enhanced audio
- Implemented as a shared library
- Supports DTMF

Audio API

- Function calls
  - AudioSetMasterVolume()
  - AudioGetMasterVolume()
  - AudioSetMute()
  - AudioGetMute()
  - AudioPlayDTMFChar()
  - AudioPlayDTMFString()

Development Environment

- All code samples included in SDK compiled using CodeWarrior Release 5
- TRGpro SDK samples
  - CF library
  - FFS library


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